Drawing to Documentation by ChatGPT [Task breakdown]

```
docs/final-implementation-plan.md
```

Below is the fully expanded version based strictly on your sketch (with slight structural improvements but no idea changes).

🧾 ERP / Business Management System

Final Implementation Planning (RDBMS Project)

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1. System Overview

This project is an **open-source Business Management & ERP system** built for the RDBMS course project.

It includes:

- Employee Management
- Warranty & Service Tracking
- System Settings
- Accounting Ledger
- Product Management
- Activity Logging

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2. Core Modules Remaining

3. Employee Module (Assigned to: Didhiti)

3.1 Features to Implement

3.1.1 Employee CRUD

- List employees
- Add employee
- Edit employee
- Delete employee

3.1.2 Employee Data Fields

Each employee must have:

- Full Name
- Phone
- Email
- NID
- Photo
- Address
- Role
- Status (Active / Inactive)

3.1.3 Access Control

- Give user access (create username)
- Role-based access system
- Permission management

3.1.4 Payroll & Salary Tracking

- Salary information

- Payment history
- Payroll status
- Salary adjustments

3.1.5 Transaction History

- Track employee-related financial transactions
- Bonus
- Salary
- Debt adjustment
- Other payments

3.1.6 Attendance Tracking

- Employee attendance system
- Daily login tracking
- Store in database

3.1.7 Reports

- Employee report
- Attendance report
- Salary report

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4. Warranty Module (Assigned to: You)

This is the most complex logic-based module.

4.1 Warranty Registration Rules

4.1.1 When Warranty Is Created

Warranty starts from:

- Sale Date

  OR

- Purchase Date

  OR

- Manufacture Date

👉 User should be able to configure which one applies.

4.2 Product Enhancements Required

4.2.1 Add Field in Product Table

Add:

```
manufacture_date
```

- Nullable
- Editable
- Available in product form

4.3 Warranty Tracking Flow

Step 1: Add Product

- System stores warranty duration
- System stores manufacture date
- System stores serial number

Step 2: Sell Product

- Create sale record
- Assign serial number

- Register warranty automatically
- Set warranty start date based on configured rule

Step 3: Product Damaged / Warranty Claim

When customer brings item:

1. Check warranty validity
2. Verify serial number
3. Confirm product is not already disposed

Step 4: Service Outcome

Possible outcomes:

- Repair & return same item
- Replace with new item
  - New item may have:
    - Fresh warranty
    - Remaining warranty
- Dispose damaged item
- Exchange with manufacturer
- Replace with free item

4.4 Serial Number Logic

- Serial number must be unique
- Serial number must be registered in DB
- Serial cannot be reused if:
  - Already disposed
  - Already replaced

4.5 Warranty Status States

Each item must have a status:

- Active
- Expired
- Claimed
- Replaced
- Disposed

4.6 Alerts

System should:

- Alert if warranty is about to expire
- Alert if expired before sale
- Special handling for perishable products (like batteries)

4.7 Accounting Integration

When replacement happens:

- Record inventory movement
- Record cost adjustment
- Reflect in accounting ledger

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5. System Settings Module (Assigned to: Anjim)

5.1 System Setup Flow

First-time setup must:

1. Create default admin user

2. Set business name
3. Configure currency
4. Configure service type

5.2 System Settings Page

Should include:

- Add / remove employee roles
- System log (view activity)
- Storage management
- Module enable/disable
- System update info
- Set default warranty rule

5.3 Activity Log

System must log:

- User login
- Data modifications
- Warranty actions
- Employee actions
- Sales actions

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6. Accounting / Ledger Module

Ledger should track:

- Carryover balance
- Debit
- Credit

- Adjustments (both credit and debit)
- Contact details

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7. Database Considerations (Very Important for RDBMS)

Make sure to include:

- Proper Primary Keys
- Foreign Keys
- Cascading rules
- Indexing for serial numbers
- Transaction handling for:
  - Sales
  - Warranty claim
  - Replacement

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8. Architecture Flow

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Product → Sale → Warranty → Claim → Replace/Repair → Accounti
ng Update
```

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9. Things That Must Not Be Missed

- Serial number uniqueness constraint
- Warranty cannot exist without sale record
- Proper foreign key relationships
- Role-based authentication

- Audit trail for academic evaluation
- Reports for each module
- Data validation everywhere

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10. Team Task Distribution

🔵 You

Warranty Module

Database logic

Warranty flow

Serial management

Replacement logic

🟢 Didhiti

Employee Module

Attendance

Payroll

Reports

🟡 Anjim

System Settings

Activity log

Role management

Initial setup flow

If you want, I can now:

- Create a more technical **database schema planning MD**
- Or create a **full implementation sprint checklist version**

- Or refine it to match your project proposal PDF strictly for grading alignment**

Tell me which direction you want.